STAT3003

Mean performs best when we have a symmetric distribution with thin tails. If skewed, use the **median**. The **mean** may be more affected by the tails.

The *p*th percentile

Case 1: If $\frac{np}{100}$ is **not an integer**: Let k = smallest integer > $\frac{np}{100}$ the pth percentile is the kth smallest sample point

Case 2: If $\frac{np}{100}$ is **an integer**: The pth percentile is the average of the $(\frac{np}{100})$th and $(\frac{np}{100}+1)$th smallest sample points

Positively Skewed / Right Skewed: Upper hinge is farther from the Median than the lower hinge. The distribution has a longer tail on the right side

Negatively Skewed / Left Skewed: Lower hinge is farther from the Median than the upper hinge. The distribution has a longer tail on the left side

Population Parameters

$\mathrm{Var}(aX+bY+c)=a^2\mathrm{Var}(X)+b^2\mathrm{Var}(Y)+2ab\mathrm{Cov}(X,Y)$

Sample Correlation Coefficient: $r=\frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{s_x}\right)\left(\frac{y_i-\bar{y}}{s_y}\right)$

$Cov(X,Y)=\frac{1}{N}\sum_{i=1}^{N}[(y_i-\mu_Y)(x_i-\mu_X)]=E(XY)-E(X)E(Y)$ $\rho_{X,Y}=\frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}$

Binomial: $X\sim B(n,p)$.If $X\sim B(n,p)$ and $np\geq 10$, $n(1-p)\geq 10$, $\mu_X=np$, $\sigma_X=\sqrt{np(1-p)}$, $Z=\frac{X-np}{\sqrt{np(1-p)}}$

Normal: $f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

corrected population variance: $S^2=\frac{1}{N-1}\sum_{i=1}^{N}(u_i-\mu)^2$

The factor $\frac{1}{N-1}$ simplifies many formulas in statistical analysis. This adjustment is useful in various statistical contexts. σ^2 better reflects the idea of variance as an expectation: $\mathrm{Var}(Y)=E[(Y-\mu)^2]$

The quantity $E[\hat{\theta}-\theta]$ is called the **bias** of the estimator $\hat{\theta}$. We want our estimators to be unbiased, i.e., to have zero bias. An unbiased estimator satisfies $E[\hat{\theta}]=\theta$, $MSE(\hat{\theta})=Var(\hat{\theta})$

$MSE(\hat{\theta})=Var(\hat{\theta})+[Bias(\hat{\theta})]^2$

Unbiased: $E[\hat{\theta}]=\theta$, **Precise**: $Var(\hat{\theta})$ is small (The estimates from different samples are close to each other), **Accurate**: unbiased&precise

Survey Terminology:

1. Population: A collection of elements about which we want to make inferences
2. Element: An element is an object on which a measurement is taken
3. Sampling Units: Partition the population
4. Frame: A frame is a list of sampling units
5. Sample: A collection of sampling units drawn from a single or from multiple frames

Probability Sampling: Each population unit has a known and non-zero probability of being selected, and the sampling process relies entirely on a random mechanism. **Can make probability statements (e.g. *Confidence Intervals*)** Randomisation balances out all factors inherent in a population

Quota Sampling: Quotas are first established based on certain characteristics (such as gender, age, region), then interviewers **subjectively** or conveniently select samples within each quota. **Can not make probability statements**

Simple Random Sampling(SRS) : Random sampling without replacement such that every possible sample of n units is equally likely 所有可能样本组合概率相等

$\pi_i=P(\text{Element } i \text{ is in the sample})$

We can write $\pi_i = E[Z_i]$ where $Z_i = 1$ if element i is included in the sample, $Z_i = 0$ otherwise
$\pi_i = \frac{\text{Number of samples containing element } i}{\text{Total number of possible samples}}$
$\pi_i = Pr(Z_j = 1) = \frac{\binom{N-1}{n-1}}{\binom{N}{n}} = \frac{\frac{(N-1)!}{(n-1)!(N-n)!}}{\frac{N!}{n!(N-n)!}} = \frac{n}{N} = E[Z_i]$

Methods

1. Draw Lots 2. Random Number Table 3.Random Numbers in Excel/R

Notation（抽样率 & f.p.c.）

- **population size** N，**sample size** n，**sampling fraction** $f = \frac{n}{N}$，**finite population correction** f.p.c.$= 1 - f = 1 - \frac{n}{N}$。
- **population mean** $\mu = \frac{1}{N}\sum_{j=1}^{N} u_j$。样本观测记为$Y_1, \ldots, Y_n$，样本均值$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$。
- 记$Z_j = \mathbf{1}\{\text{单位 } j \text{被抽中}\}$，则$\bar{Y} = \frac{1}{n}\sum_{j=1}^{N} u_j Z_j$。

sample mean μ

$\bar{Y} = \frac{1}{n}\sum_{j=1}^{N} u_j Z_j$, u_j is the value of the j-th population element, Z_j equals 1 if element j is in the sample, 0 otherwise

$E[\bar{Y}] = \frac{1}{n}\sum_{j=1}^{N} u_j E[Z_j] = \frac{1}{n}\sum_{j=1}^{N} u_j \frac{n}{N} = \frac{1}{N}\sum_{j=1}^{N} u_j = \mu$

calculate $Var(\bar{Y})$:

For $U := \sum_{i=1}^{n} a_i X_i$, we have: $Var(U) = \sum_{i=1}^{n} a_i^2 Var(X_i) + \sum_{i=1}^{n}\sum_{j\neq i} a_i a_j Cov(X_i, X_j)$

Applying this to our sample mean:

$Var(\bar{Y}) = \sum_{j=1}^{N}\left(\frac{u_j}{n}\right)^2 Var(Z_j) + \sum_{j=1}^{N}\sum_{k\neq j}\frac{u_j}{n}\frac{u_k}{n} Cov(Z_j, Z_k)$

$Var(Z_j) = \frac{n}{N}\left(1 - \frac{n}{N}\right), E(Z_j) = \frac{n}{N}$

$Cov(Z_j, Z_k) = E[Z_j Z_k] - E[Z_j]E[Z_k] = E[Z_j Z_k] - \left(\frac{n}{N}\right)^2$

Note that $Z_j Z_k = 1$ only when both elements j and k are in the sample, otherwise it equals 0.
$E[Z_j Z_k] = \frac{\binom{N-2}{n-2}}{\binom{N}{n}} = \frac{n(n-1)}{N(N-1)}$
$Cov(Z_j, Z_k) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n(N-n)}{N^2(N-1)} = -\frac{n(1-\frac{n}{N})}{N(N-1)}$
$Var(\bar{Y}) = \frac{N-n}{N-1}\frac{1}{n}\sigma^2$
$\widehat{Var}(\bar{Y}) = \frac{N-n}{N-1}\frac{\hat{\sigma}^2}{n} = \left(1 - \frac{n}{N}\right)\frac{\hat{\sigma}^2}{n}$

Where $\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$ is the unbiased estimator of the population variance.

$\bar{Y} \pm z_{1-\alpha/2}\sqrt{\frac{N-n}{N}}\frac{\hat{\sigma}}{\sqrt{n}}$， $\bar{y} \pm z_{1-\alpha/2}\sqrt{\frac{N-n}{N}}\frac{s}{\sqrt{n}}$

t-CI for μ: $\bar{y} \pm t_{n-1,1-\alpha/2}\sqrt{\frac{N-n}{N}}\frac{s}{\sqrt{n}}$， $= \bar{y} \pm t_{n-1,\alpha/2}\sqrt{\frac{1-\frac{n}{N}}{n}\cdot\frac{1}{n-1}\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2}$.

Sample size for μ（目标半宽d）：$n = \frac{1}{\frac{1}{N} + \frac{d^2}{v^2 t^2_{n-1,1-\alpha/2}}}$

Initial approximation: Start with the z-distribution:$n_0 = \frac{1}{\frac{1}{N} + \frac{d^2}{v^2 z^2_{1-\alpha/2}}}$$

Iteration: Use the formula recursively: $n_{i+1} = \frac{1}{\frac{1}{N} + \frac{d^2}{v^2 t^2_{n_i-1,1-\alpha/2}}}$

We choose the first n which is larger than the right hand side of the equation
Estimating a total $\tau = N\mu$, $\tau = \sum_{j=1}^{N} u_j$, where: u_j is the value of the j-th unit in the population

$Var(\hat{\tau}) = N^2 \cdot \frac{N-n}{N-1} \cdot \frac{\sigma^2}{n} = \frac{N-n}{N-1} \cdot \frac{N^2}{n} \cdot \sigma^2$

$\widehat{Var}(\hat{\tau}) = N^2 \cdot (1 - \frac{n}{N}) \cdot \frac{\hat{\sigma}^2}{n}$

$\hat{\tau} \pm t_{n-1,1-\alpha/2} \cdot N \cdot \sqrt{(1 - \frac{n}{N}) \cdot \frac{\hat{\sigma}^2}{n}}$ = $N\bar{y} \pm t_{n-1,\alpha/2}\, N \sqrt{\dfrac{1-\frac{n}{N}}{n} \cdot \dfrac{1}{n-1} \sum_{i=1}^{n} \left(y_i - \bar{y}\right)^2}$

Sample size for μ（目标半宽d）: $n = \frac{1}{\frac{1}{N} + \frac{d^2}{N^2 v^2 z^2_{1-\frac{\alpha}{2}}}} \cdot (1 - \frac{1}{N})$

$n = \frac{1}{\frac{1}{N} + \frac{d^2}{N^2 v^2 t^2_{n-1,1-\alpha}}}$

Estimating a proportion p

$p = \frac{1}{N} \sum_{j=1}^{N} y_j$. That is, p is a special case where $\hat{p} = \frac{1}{n} \sum_{i=1}^{n} Y_i$, where $Y_i = 1$ if element i has the characteristic and $Y_i = 0$ otherwise

$\Rightarrow Var(\hat{p}) = \frac{N-n}{N-1} \frac{1}{n} p(1-p)$

$\widehat{\text{Var}}(\hat{p}) = \dfrac{1 - \frac{n}{N}}{n}\, \hat{p}(1-\hat{p})$

When the Y_i are Bernoulli random variables, $\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2 = \frac{n}{n-1} \hat{p}(1-\hat{p})$

$\hat{p} \pm \sqrt{(1 - \frac{n}{N}) \frac{1}{n-1} \hat{p}(1-\hat{p})} \cdot t_{n-1,1-\frac{\alpha}{2}}$, Because $\sigma^2 = p(1-p)$ will be unknown. If it is known we would know p.

What is large enough, in this case? We require $n\hat{p} \geq 5$ and $n(1-\hat{p}) \geq 5$

$n = \frac{1}{\frac{1}{N} + \frac{d^2}{p(1-p) z^2_{1-\frac{\alpha}{2}}}} (1 - \frac{1}{N})$

If we don't know p: For a conservative sample size, use $p = 0.5$. ($p(1-p)$ is maximized at $p = 0.5$)

Stratified Random Sampling

Reduction of estimator variance,Reduction of cost, Comparison among sub-groups, Prevent really bad (unrepresentative) samples

L = Number of strata

N_i = Number of sampling units within stratum i

N = Number of sampling units in population, so $N = N_1 + N_2 + ... + N_L$

$\bar{Y}_{st}$ = Estimator of population mean from stratified sampling

A subscript of i will mean that the parameter/estimator concerned is from stratum i.

E.g. $\bar{Y}_i$, μ_i, s_i^2, τ_i etc.

Estimating a total $\tau = \sum_{i=1}^{L} N_i \bar{Y}_i$

$Var(\hat{\tau}_{st}) = Var\left(\sum_{i=1}^{L} N_i \bar{Y}_i\right) = \sum_{i=1}^{L} Var(N_i \bar{Y}_i) = \sum_{i=1}^{L} N_i^2 \frac{\sigma_i^2}{n_i} \frac{N_i - n_i}{N_i - 1}$

$\widehat{Var}(\hat{\tau}_{st}) = \sum_{i=1}^{L} N_i^2 \frac{1}{n_i} \frac{N_i - n_i}{N_i - 1} \left(1 - \frac{1}{N_i}\right) \hat{\sigma}_i^2 = \sum_{i=1}^{L} N_i^2 \left(1 - \frac{n_i}{N_i}\right) \frac{\hat{\sigma}_i^2}{n_i}$

$\hat{\tau}_{st} \pm t_{df,1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\tau}_{st})}$

Intuitively, we might guess $df = df_1 + df_2 + ... + df_L$ where $df_k = n_k - 1$, the degrees of freedom from the Simple Random Sampling (SRS) in stratum k. However, this is correct only if the variances of all the strata are the same

Satterthwaite's Approximation provides a solution:

$df_{\chi'} \approx \frac{\left(\sum_{i=1}^{n} k_i \hat{\sigma}_i^2\right)^2}{\sum_{i=1}^{n} \frac{(k_i \hat{\sigma}_i^2)^2}{n_i - 1}}$, where $\hat{\sigma}_i^2$ has sample size n_i.

$df \approx \frac{\left(\sum_{h=1}^{L} k_h s_h^2\right)^2}{\sum_{h=1}^{L} \frac{(k_h s_h^2)^2}{n_h - 1}}$, where $k_h = \frac{N_h(N_h - n_h)}{n_h}$, ${s_h}^2 = \frac{n_h}{n_h - 1}\hat{p}_h(1 - \hat{p}_h)$

If the strata sample sizes are at least 30, we can avoid calculating df by using the normal distribution instead: $\hat{\tau}_{st} \pm z_{1-\frac{\alpha}{2}}\sqrt{Var(\hat{\tau}_{st})}$

Sample size for μ（目标半宽d）：$n = \frac{\sum_{i=1}^{L} N_i^3 \frac{1}{a_i} \frac{\sigma_i^2}{N_i - 1}}{d^2/z_{1-\frac{\alpha}{2}}^2 + \sum_{i=1}^{L} N_i^2 \frac{\sigma_i^2}{N_i - 1}}$

$\bar{Y}_{st} = \frac{1}{N^2} \sum_{i=1}^{L} N_i^2 \frac{\sigma_i^2}{n_i} \frac{N_i - n_i}{N_i - 1}$

$Var(\bar{Y}_{st}) = \sum_{i=1}^{L} Var(N_i \bar{Y}_i)$

$\widehat{Var}(\bar{Y}_{st}) = \frac{1}{N^2} \sum_{i=1}^{L} N_i^2 \left(1 - \frac{n_i}{N_i}\right) \hat{\sigma}_i^2$

$\bar{y}_{st} \pm z_{1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{st})}$, $\bar{y}_{st} \pm t_{df, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\bar{y}_{st})}$

$df \approx \frac{\left(\sum_{h=1}^{L} k_h s_h^2\right)^2}{\sum_{h=1}^{L} \frac{(k_h s_h^2)^2}{n_h - 1}}$, where $k_h = \frac{N_h(N_h - n_h)}{N^2 n_h}$, ${s_h}^2 = \frac{n_h}{n_h - 1}\hat{p}_h(1 - \hat{p}_h)$

Sample size for μ（目标半宽d）：$n = \frac{\sum_{i=1}^{L} \frac{N_i^3}{a_i} \frac{\sigma_i^2}{N_i - 1}}{N^2 d^2/z_{1-\frac{\alpha}{2}}^2 + \sum_{i=1}^{L} N_i^2 \frac{\sigma_i^2}{N_i - 1}}$

$\hat{p}_{st} = \frac{1}{N} \sum_{i=1}^{L} N_i \hat{p}_i$

$Var(\hat{p}_{st}) = \frac{1}{N^2} \sum_{i=1}^{L} N_i^2 Var(\hat{p}_i) = \frac{1}{N^2} \sum_{i=1}^{L} N_i^2 \cdot \frac{N_i - n_i}{N_i - 1} \cdot \frac{p_i(1-p_i)}{n_i}$

$\widehat{Var}(\hat{p}_{st}) = \frac{1}{N^2} \sum_{i=1}^{L} N_i^2 \cdot (1 - \frac{n_i}{N_i}) \cdot \frac{\hat{p}_i(1-\hat{p}_i)}{n_i - 1}$

$\hat{p}_{st} \pm z_{1-\alpha/2}\sqrt{\hat{Var}(\hat{p}_{st})}$

$\hat{p}_{st} \pm t_{df, 1-\alpha/2}\sqrt{\hat{Var}(\hat{p}_{st})}$, $df \approx \frac{(\sum_{h=1}^{L} k_h s_h^2)^2}{\sum_{h=1}^{L} \frac{(k_h s_h^2)^2}{n_h - 1}}$, Where: $k_h = \frac{N_h(N_h - n_h)}{N^2 n_h}$

$\frac{N_i}{N_i - 1} \approx 1$ (which is reasonable for large strata), the formula becomes:
$n = \frac{\sum_{i=1}^{L} N_i^2 \frac{1}{a_i} p_i(1-p_i)}{N^2 d^2/z_{1-\alpha/2}^2 + \sum_{i=1}^{L} N_i p_i(1-p_i)}$
$p_i = 0.5$ for all strata, we get: $n = \frac{\sum_{i=1}^{L} N_i^2 \frac{1}{a_i} \cdot \frac{1}{4}}{N^2 d^2/z_{1-\alpha/2}^2 + \frac{N}{4}}$
With proportional allocation ($a_i = \frac{N_i}{N}$), the formula simplifies to: $n = \frac{N^2}{4N^2 d^2/z_{1-\alpha/2}^2 + N}$
For a 95% confidence interval, $z_{1-\alpha/2}^2 = z_{0.975}^2 \approx 1.96^2 \approx 3.84$, giving: $n = \frac{N}{1.04 d^2 N + 1}$
When N is very large, the formula further simplifies to: $n = \frac{1}{1.04 d^2}$

The relative size of each stratum in the population affects how much information we gain from sampling
The variability of observations within each stratum (measured by σ_i) significantly impacts required sample sizes
The cost of obtaining observations may differ substantially between strata

- Total population size: N
- Population size of stratum i: N_i
- Total sample size: n

- Sample size for stratum i: n_i

Neyman Allocation: $n_h = n \cdot \frac{N_h \sigma_h}{\sum_{i=1}^{L} N_i \sigma_i}$

The proportional allocation formula is: $n_i = \frac{n \cdot N_i}{N}$

$a_1 = \frac{N_1}{N}, a_2 = \frac{N_2}{N}, \ldots, a_L = \frac{N_L}{N}$

We choose the a_i that minimize the variance of our estimator, $Var(\bar{Y}_{st})$

-Minimize: $\frac{1}{N^2} \sum_{i=1}^{L} N_i^2 \frac{\sigma_i^2}{n_i} \frac{N_i - n_i}{N_i - 1}$

-over all n_1,n_2,...,n_L

-Subject to: $\sum_{i=1}^{L} n_i = n$

use Lagrange Multipliers:

-$L(n_1, \ldots, n_L, \lambda) = \frac{1}{N^2} \sum_{i=1}^{L} N_i^2 \frac{\sigma_i^2}{n_i} \frac{N_i - n_i}{N_i - 1} - \lambda \left(n - \sum_{i=1}^{L} n_i \right)$

Taking the partial derivative with respect to n_i:

-$\frac{\partial L}{\partial n_i} = \frac{1}{N^2} \frac{\partial}{\partial n_i} \left[N_i^2 \frac{\sigma_i^2}{n_i} \frac{N_i - n_i}{N_i - 1} \right] + \lambda$

-$\frac{\partial L}{\partial \lambda} = \sum_{i=1}^{L} n_i - n = 0$

$$\Rightarrow n_i = n \cdot \frac{\sqrt{\frac{N_i}{N_i - 1}} N_i \sigma_i}{\sum_{j=1}^{L} \sqrt{\frac{N_j}{N_j - 1}} N_j \sigma_j}$$

Choose the ni that minimize $Var(\bar{Y}_{st})$
subject to the constraint that $c_0 + c_1 n_1 + c_2 n_2 + \ldots + c_L n_L = c$

$$\Rightarrow n_i = \frac{(c - c_0) \cdot \sqrt{\frac{N_i}{N_i - 1}} \frac{N_i \sigma_i}{\sqrt{c_i}}}{\sum_{j=1}^{L} \sqrt{\frac{N_j}{N_j - 1}} N_j \sigma_j \sqrt{c_j}}$$

当各层均值差异小而方差差异大的时候，按层抽样不起作用。

Cluster Sampling: A cluster sample is a probability sample in which each sampling unit is a collection, or cluster, of elements (to keep the cost of survey down)

We sample from all the strata, but only some of the clusters – We take measurements from only some elements in the stratum (we perform a simple random sample in each stratum) but we measure every element in the cluster

The difference between strata should be large and the differences within small. The differences within each cluster should reflect the whole population, but the differences between should be small.

N: The total number of clusters in the population.
n: The number of clusters selected in the sample.
M_i: The number of elements in cluster i.
M: The population size, i.e. $M = \sum_{i=1}^{N} M_i$.
m_i: The sample size in the i-th cluster.
$\bar{M}$: The average cluster size in the population, $\bar{M} = \frac{M}{N}$.
Y_{ij}: The j-th observation from the i-th cluster.
Y_i: The total of observations in the i-th cluster, $Y_i = \sum_{j=1}^{M_i} Y_{ij}$.

Estimating τ

The population total is $\tau = \sum_{i=1}^{N} \sum_{j=1}^{M_i} y_{ij} = \sum_{i=1}^{N} y_i$

Estimator $\hat{\tau} = N\frac{1}{n}\sum_{i=1}^{n} Y_i = N\bar{Y}$

Let μ_c be the average total for a cluster, that is: $\mu_c = \frac{1}{N}\sum_{i=1}^{N} Y_i$

$\hat{\tau}$ is unbiased.

$\mathrm{Var}(\hat{\tau}) = N^2\,\mathrm{Var}(\bar{Y}) = N^2\,\frac{N-n}{N-1}\,\frac{1}{n}\,\sigma_c^2$, where σ_c^2 is the variance of the cluster totals.

$\widehat{\mathrm{Var}}(\hat{\tau}) = N^2\left(1-\frac{n}{N}\right)\frac{\hat{\sigma}_c^2}{n}$, where $\hat{\sigma}_c^2$ is the sample variance of the cluster totals.

$100(1-\alpha)$ 的C. I. $\Rightarrow\ \hat{\tau} \pm t_{n-1,\,1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$

簇数的样本量估算 We may find an appropriate number of clusters to sample to achieve a 100(1−α)% C.I. of width $2d$ by solving $d = z_{1-\alpha/2}\sqrt{\mathrm{Var}(\hat{\tau})}$ for n.

Note: this will depend on σ_c^2, which is almost certainly unknown. Plug in a reasonable guess for σ_c^2 to obtain a number.

Two stage cluster sampling

Advantages 1. More practical when a complete sampling frame is unavailable 2. Reduces costs when sampled elements are geographically dispersed

Implementation Process

Large clusters: Tend to be heterogeneous $\Rightarrow$ Require larger samples within each cluster $\Rightarrow$ Can use fewer clusters overall

Small clusters: Tend to be homogeneous $\Rightarrow$ Require smaller samples within each cluster $\Rightarrow$ Need more clusters to achieve precision

Notion: 与之前一样除了

m_i = number of elements selected from the i-th cluster by SRS

Y_{ij} = the j-th observation in the sample from cluster i

Hence, the sample mean for cluster i is $\bar{Y}_i = \frac{1}{m_i}\sum_{j=1}^{m_i} Y_{ij}$. (在整群分类中是$n_i$)

A SRS unbiased estimator for the total of cluster i is $\hat{Y}_i = M_i\,\frac{1}{m_i}\sum_{j=1}^{m_i} Y_{ij}$.

Therefore, an unbiased estimator of the population total $\tau = \sum_{i=1}^{N} Y_i$ is $\hat{\tau} = N\,\frac{1}{n}\sum_{i=1}^{n}\hat{Y}_i$.

Estimating

对于一个two-stage我们要通过两个指标来看一个Estimator到底是好还是不好

Conditional expectations 和 Conditional variances

Let $\hat{\theta}$ be an estimator based on a two-stage sample, s_1 be the (random) set of primary sampling units selected in the first stage.

Then by the **Conditional expectations**, $E\left[E(\hat{\theta}\mid s_1)\right] = E(\hat{\theta})$.

And by the **Conditional variances**, $\mathrm{Var}(\hat{\theta}) = \mathrm{Var}\left(E[\hat{\theta}\mid s_1]\right) + E\left[\mathrm{Var}(\hat{\theta}\mid s_1)\right]$.

$[\mathrm{Var}(X|Y=y) = E[(X-E[X|Y=y])^2|Y=y]]$

Example of τ

For our estimator $\hat{\tau}$

1. $E[\hat{\tau}|s_1] = N\frac{1}{n}\sum_{i=1}^{n} E[\hat{y}_i|s_1]$. Since the $\hat{y}_i$ are SRS estimators of the cluster totals y_i, $E[\hat{y}_i|s_1] = Y_i \Rightarrow E[\hat{\tau}|s_1] = N\frac{1}{n}\sum_{i=1}^{n} Y_i$

2. Hence $E[\hat{\tau}] = E[E[\hat{\tau}|s_1]] = E\left[N\frac{1}{n}\sum_{i=1}^{n} Y_i\right] = NE[\bar{Y}] \Rightarrow E[\hat{\tau}] = \tau$, so our estimator is unbiased.

$Var(\hat{\tau}) = N^2\frac{N-n}{N-1}\frac{\sigma_c^2}{n} + \frac{N}{n}\sum_{i=1}^{N} M_i^2\frac{M_i-m_i}{M_i-1}\frac{\sigma_i^2}{m_i}$, where σ_c^2 is the population variance of the cluster totals, i.e. the y_is and σ_i^2 is the population variance within the i-th cluster

$\widehat{Var}(\hat{\tau}) = N(N-n)\frac{1}{n}\hat{\sigma}_c^2 + \frac{N}{n}\sum_{i=1}^{n} M_i(M_i-m_i)\frac{1}{m_i}\hat{\sigma}_i^2$, where $\hat{\sigma}_c^2$ is the sample variance of the estimated cluster totals (the $\hat{Y}_i$) and $\hat{\sigma}_i^2$ is the sample variance inside cluster i

Thus the $100(1-\alpha)$ C.I. for τ will be $\hat{\tau} \pm Z_{1-\alpha/2}\sqrt{\widehat{Var}(\hat{\tau})}$

Estimating Population Proportion p

$\hat{p} = \frac{N}{M}\frac{1}{n}\sum_{i=1}^{n}\hat{Y}_i$, where $\hat{Y}_i$ is the estimator for the number of people in cluster with the characteristic

We may write $\hat{Y}_i = M_i\hat{p}_i$ to relate p to the sample proportions inside the clusters

Unbiased estimator for the variance is

$\widehat{Var}(\hat{p}) = \frac{1}{M^2}N(N-n)\frac{1}{n}\hat{\sigma}_c^2 + \frac{1}{M^2}\frac{N}{n}\sum_{i=1}^{n}M_i(M_i-m_i)\frac{1}{m_i}\hat{\sigma}_i^2$

Recall: $\hat{\sigma}_i^2$ is the estimator for population variance of cluster i, hence $\hat{\sigma}_i^2 = \frac{m_i}{m_i-1}\hat{p}_i(1-\hat{p}_i)$

$\hat{\sigma}_c^2$ is the sample variance of the estimated cluster totals, hence $\hat{\sigma}_c^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(M_i\hat{p}_i - \frac{M}{N}\hat{p}\right)^2$

assumption: officially we need: n, N, N-n shall be large, but if Y_i can be assumed as i.i.d normal, we can use CI formula.

officially we need: n, N, N-n shall be large (which is satisfied), Y_i shall be roughly i.i.d normal (which is not satisfied since Y_i is either 2 or -1). But since n/N/N-n are large, sample variance is a good estimate of population variance, so using z quantile under the large n/N/N-n condition is approximately ok.